

Questions for Leslie Lamport

A PhooD Research Seminar Q&A

Prepared by PhooD organizers and attendees, Michtom School of Computer Science

The following seventeen questions were prepared and asked during a Q&A with Turing Award laureate Leslie Lamport, hosted by the PhooD research seminar series at Brandeis University on April 9, 2025.

01

How might formal methods (and TLA+ in particular) be applied or adapted to help ensure correctness, reliability, or even fairness in large-scale AI pipelines, given the probabilistic and often opaque nature of current AI-driven approaches? Do you see a feasible path for integrating rigorous specification and reasoning into systems that revolve around inherently ambiguous human language?

02

What kinds of systems are not suitable for formal/mathematical thinking?

03

In your experience, how effective is AI at designing systems and algorithms compared to humans?

04

With regards to thinking about programming from a mathematical standpoint, you've said that the "mathematical education in this country is pretty terrible". You mentioned that many people, including senior programmers, are afraid of mathematics. What should we do differently? How do you think we should teach people to think more mathematically?

05

In order to conduct creative work, do you believe it is more meaningful to pursue an academic position or to enter industry?

06

How do you view the challenges of pursuing innovation in academia, given the growing reliance on large teams and significant resources as seen in industry?

07

How can one avoid becoming just a “cog in the machine” when starting in a junior position in the industry? What roles and teams allow you to maximize your impact?

08

How do the earliest companies you’ve worked for (Massachusetts Computer Associates, DEC, etc.) compare to the current tech giants of today?

09

What fundamental, unresolved challenges in traditional systems research merit continued attention?

10

What emerging areas in computer science are you paying attention to?

11

Throughout your career, you have seen multiple transformations in computing—from centralized to distributed cloud systems, and now to specialized hardware (e.g. ASICs) and generative AI-driven architectures. In your experience, what key fundamental challenges have re-appeared across these different technology advancements?

12

You are the sole author on many of your papers. How do you stay sane when working alone?

13

How do you choose what problems to work on?

14

You’ve said “Coding requires Thinking”, and “Thinking requires writing”. What separates a productive writing process from a non-productive writing process?

15

How do you approach communicating complex ideas when you write?

16

What was Brandeis like in the 70s when you were in graduate school?

17

What were the initial challenges you faced in creating LaTeX? Did you anticipate it becoming the standard for technical document preparation?